Unit 3 · Lesson 10 Electron Configurations II

Ions, shorthand, valence, Lewis dots Name _____ Per _____

Keywords: ion, cation, anion, noble-gas shorthand, core electrons, valence electrons, Lewis dots, isoelectronic. *Pencil. Fix wrong answers in a second color. Printed text is yours to read; every blank is a decision.*

Question of the day

Do before the bell rings; revisit at the end

Why doesn't your lunch blow up? Circle one: **A** It isn't really potassium. **B** The potassium already lost an electron. **C** Too little of it to react. **D** The peel keeps the water out.

My reason: _____ After the lesson I would change to: _____

Retrieve

5 min, no notes

1. Config of P (Z = 15): _____ **2.** Electrons in a d subshell: _____ **3.** 17 p^+, 18 e^-. Charge: _____

Predict, then watch

commit before the animation

Argon has 18 electrons: $1s^2 2s^2 2p^6 3s^2 3p^6$. Potassium has one more. Electron 19 goes to (circle): **A** 3d **B** 4s **C** 4p. Because _____

After watching: electron 19 went to _____; the outermost shell holds ____ electron.

Ions and shorthand

An **ion** is an atom that has gained or lost electrons. **Protons never change**; electrons leave from, and join, the **outermost shell** (the highest *n*). A **cation** lost electrons (positive). An **anion** gained electrons (negative). **Shorthand** swaps the core for the noble gas that comes **before** the element: Na = $[Ne]3s^1$, not $[Ar]3s^1$.

K^+: protons ____ electrons ____ charge ____ Shorthand for Al (Z = 13): _____

Core vs valence: gallium, Z = 31

predict first: how many valence electrons does gallium have? My guess: ____

Valence electrons are in the outermost shell (highest *n*). Everything inside is **core**. The 3d fills after 4s, but its shell number is 3.

Shell *n*	Subshells in that shell	Electrons	Circle one
1	$1s^2$	_____	core valence
2	$2s^2 2p^6$	_____	core valence
3	$3s^2 3p^6 3d^{10}$	_____	core valence
4	$4s^2 4p^1$	_____	core valence

Ga has ____ valence electrons.

Check: do the shell counts add to 31? _____

Lewis dots

atoms only: no brackets or charges yet

Dots show **valence electrons only**. Place the s pair on **top**, then one dot each on the **right, bottom, left**, then pair up. Na and Mg are drawn; do Al to Ar. P should get ___ dots.

Na	Mg	Al	Si	P	S	Cl	Ar
1 valence	2 valence	___ valence	___ valence	___ valence	___ valence	___ valence	___ valence

Stretch

finished early? start here

Stretch 1 Draw the orbital boxes for N (Z = 7). Unpaired electrons: ____ Full configuration of Ca: _____

Stretch 2 S^{2-}, Cl^-, K^+, Ca^{2+} and Ar share one configuration. Protons in each: ___, ___, ___, ___, ___. What differs? _____

Stretch 3 Fe is $[Ar]3d^6 4s^2$. Write Fe^{2+}: _____ and Fe^{3+}: _____ Which electrons left first? _____

Worked examples: I do, we do, you do

3 marks each

I do: Mg^{2+}

1. Atom has 12 electrons: [Ne]$3s^2$
2. Lose 2 from $n = 3$:
[Ne]$3s^2 \rightarrow$ [Ne]
3. Same as neon: Mg^{2+} = [Ne]

Protons stay 12.

We do: Cl^-

1. Electrons in the atom: ____
2. Gain 1: [Ne]$3s^2$3p__
3. Matches: [____]

Cl^- =_____

You do: S^{2-}

1. Electrons: _____
2. Full: _____
3. Shorthand: _____

Two ways to get 18. Which did you use?

Pair trade

write your four, swap, check with the table

Do the four ions in your letter's rows. Then check your partner's four. Disagree? Find the **first** line where you differ.

	Ion	Electrons	Shorthand	Partner OK?
A	N^{3-}			☐
A	Li^+			☐
A	K^+			☐
A	Br^-			☐
B	Al^{3+}			☐
B	O^{2-}			☐
B	Ca^{2+}			☐
B	Se^{2-}			☐

Br^- and Se^{2-} hide the $3d^{10}$ trap: is the 3d inside the bracket?

Error hunt

name the flaw, then fix it

	Claim	What is wrong?	Fix
A	Na = [Ar]$3s^1$		
B	Ca^{2+} = [Ar]$4s^2$		
C	Gallium has 13 valence electrons.		
D	Sulfur's Lewis symbol has 16 dots.		

Hinge: arsenic, [Ar]$3d^{10}4s^24p^3$

whiteboards up on three

How many valence electrons? **A** 3 **B** 5 **C** 15 **D** 33 My board: ____ Right answer: ____ If I missed it, my error was: _____

Open questions

no wrong thinking here: argue it

Do Lewis dots work for d electrons? What breaks?

_____
_____

4s fills before 3d. Why does it leave first?

_____
_____

Exit ticket

tear off and hand in

1. Ca^{2+} shorthand: _____ **2.** One particle with the same configuration: _____

3. Draw the Lewis dots for phosphorus in the box. Valence electrons: ____



___ valence